

Problem Set 1 EDPs October 2025

1. Solve $y' = \frac{y}{\sqrt{x+1}}$ with $y(3) = 1$.
2. Solve $y' - 2xy = x$ with $y(0) = 0$.
3. Solve $(2y + 2xy)y' + y^2 + 1 = 0$ with $y(0) = 1$
4. Consider the following eigenvalue problem:

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + \lambda y = 0$$

with $y(0) = 0$, $y(1) = 0$

Find all possible values for λ and the corresponding eigenfunctions.

5. Modify the code `FiniteDifferenceIIIEigens2025.ipynb` to find the eigenvalues/eigenvectors of the discrete problem corresponding to problem 1.
6. Using `FiniteDifferenceIVNonConstCoeffs2025.ipynb` as a starting point, write some code to find numerical solutions to Cauchy-Euler equations of the form $x^2 \frac{d^2y}{dx^2} + ax \frac{dy}{dx} + cy = 0$, with boundary conditions $y(1) = 2$, $y(2) = 8$.

(Find the real solution to $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} + 3y = 0$ with $y(1) = 2$, $y(2) = 8$ and compare the real solution with the numerical one.)
7. Use separation of variables to find possible solutions of the following:

$$tu_t = u_{xx} + 2u \text{ with boundary conditions } u(0, t) = u(\pi, t) = 0$$

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